Tufts University

Sundial Plant Hanger



By: Breoghan Conchas Marquez, Iris Guzman De Leon, Amanda Moore-Rinehart, and Raphael Lemus Solis

The project's primary purpose was to create a sundial–a timekeeping device that uses the shadow of a stick cast by the sun to tell the time. However, since we wanted to create a vertical sundial, we thought it would also be a good idea to make it into a hook to hold a plant. The concept of the project originated from the fact that something could be made out of pewter–a metal–also known as bismuth tin. It was something that we used for a lab in Materials and Manufacturing. We thought pewter would be durable enough to hold a plant as well. Therefore, the sundial plant hanger came about, which will be able to hold a plant and tell the time. Not only did we use pewter, but we used aluminum and acrylic to make the board. The acrylic was used mainly for aesthetics, while the aluminum was cut with the waterjet cutter to make the gnomon's shadow readable. Every piece was cleaned up with a Dremel, and to piece everything together, we drilled holes and used the tap wrench onto the acrylic to keep the screws in place.

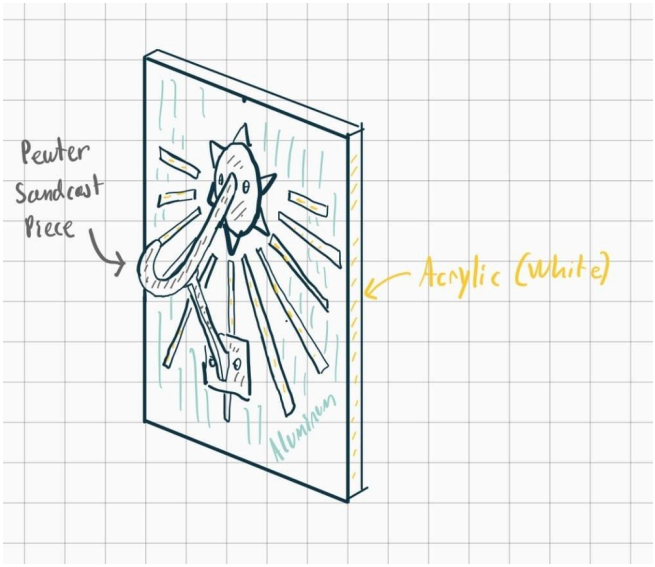


Figure 1: The final design of our project

When we first set out to create the product, we had some meaningful intentions to get the most out of the project. We wanted to create a multi-functional device to learn how to properly sand-cast–a technique used to make parts using liquefied metal and sand as a mold–without too many defects. We also wanted to create 3D model objects using SolidWorks to help us build and make an FEA model–a way to tell how much weight the design could withstand without breaking–of the device.

We created multiple prototypes to ensure our idea would come out as we wanted. First, we started a prototype made out of cardboard to know the gnomon's position–the sundial's stick. Then, we created many 3D models and printed them out to see what they looked like. We developed the idea further into the final pieces. However, the pieces were larger than the mold in the workshop, so we had to create two molds for the two pieces we needed to sand cast. Along with 3D modeling of the hook and sun, we developed the aluminum board design to tell the time.



Figure 2: The final 3D part to create the hook

Everything did not go as smoothly as we wanted. Many setbacks took us a couple of days to finally overcome the challenge. The first one was the mold box for the sand casting, which broke because it was not sturdy enough. As we mentioned, we did multiple 3D prints to ensure that sand casting would work. The first time we did it, it did not print how we wanted it to. The second time we did it, we realized that the whole thing needed a different shape. The third time, we realized that the way the sun was before would lead to some errors in the mold. Therefore, we had to combine the hook with a circle part of the sun with the hook and the design of the sun separately. We faced problems with the sun, and the hook was also troubling because of its cylindrical shape. Therefore, we had to split the 3D model of the hook in half and connect them with a piece to get the cylindrical shape, so we had to scale it to the correct size, covering 12. We also had to scale down the board's design to get the metal to fit in the machine we would cut, and during that time, the machine was having some issues.

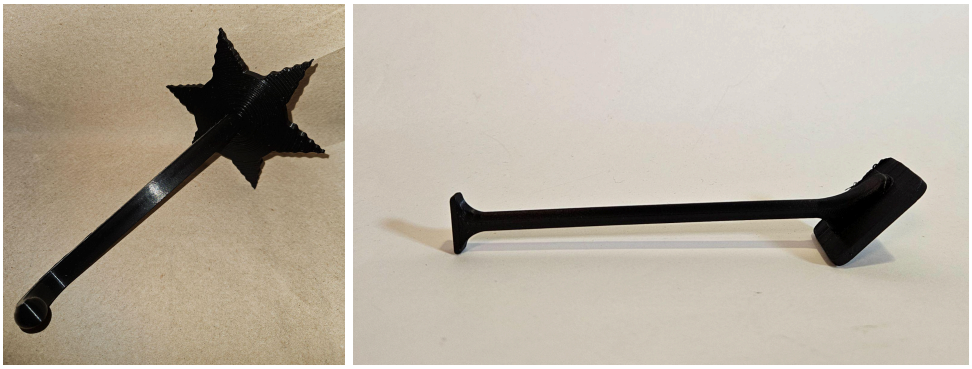


Figure 3: Failed 3D print for our initial design to sand-cast

One of our successes was correctly sand-casting the hook. We worked on it for weeks because the initial design contained errors. However, after overcoming those obstacles and working together as a team, we finally reached a shape we were happy with. Another was creating an FEA model of the object since pewter is known to be brittle; however, the FEA model could tell us how much weight it could withstand before breaking.



Figure 4: Images of how everything turned out in the end

Our design can hold 13.97 kg or about 30.79 lbs. Therefore, the goal of holding a 10 lb flower pot is doable as long as no extra weight is put on the hook. It changes the x-direction by 0.402 mm for the maximum amount of 10 lbs of force. The most challenging part of the project was learning how to sand cast a cylindrical item.

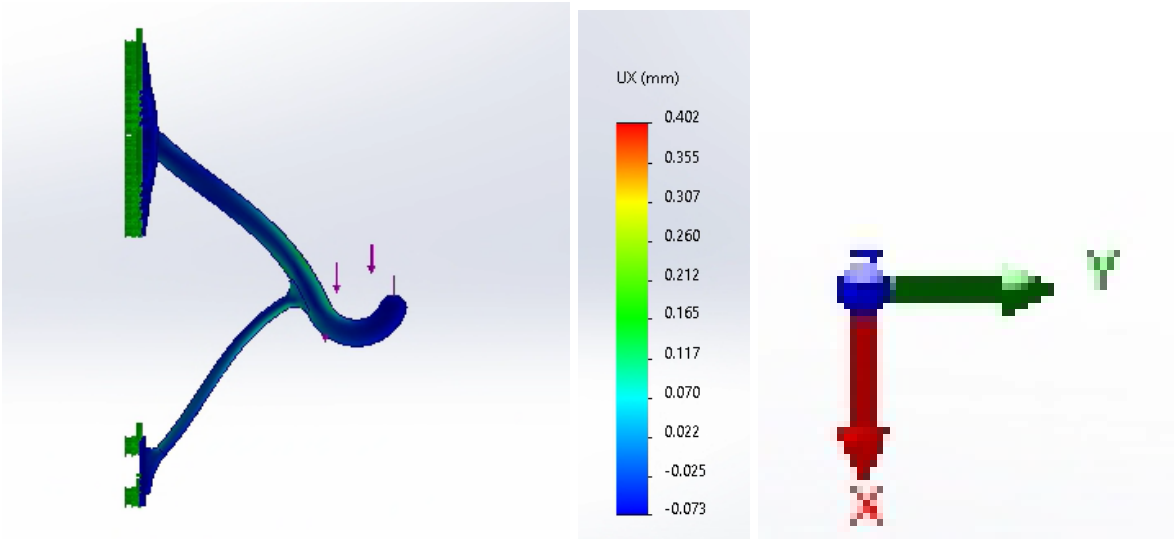


Figure 5: This is the FEA model for 45 Newtons, meaning it could hold 10 lbs securely

The design constraint is that we cannot make it bounded to every wall, meaning it is only south-bounded. This is due to the sun's positioning and how the shadow cast on the gnomon differs if the sundial is placed on a south and southwest-bounded wall. The difference between the two is the positioning of the gnomon. When it is southbound, the gnomon faces directly at 12; however, for southwestbound, the gnomon faces in a different direction. Therefore, if we were to do the project again, we would figure out a way for the gnomon to determine the right bound for whichever wall it is on.



Figure 6: The final product